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METRIC UNFOLDING: DATA REQUIREMENT FOR UNIQUE SOLUTION & CLARIFICATION OF SCHÖNEMANN'S ALGORITHM[1]

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The object of this paper is to clarify Schönemann's unfolding algorithm and, in particular, to make it clear that the equations numbered (3.2) in Schönemann's [1970] article, which define Schönemann's solutions, are not a complete set of restraints for the purpose of defining metric unfoldings. Namely, Schönemann has transformed the original equations which define an unfolding to a set of linear and non-linear equations of which he uses only the linear equations to define his solutions. Given infallible data (solution(s) exist) Schönemann's solutions will include the correct solutions. If enough data are available so that there are enough linear equations to uniquely determine a single solution, then Schönemann's solution will coincide with the correct solution. Let P and Q denote the number of elements in the two sets of points, the interset distances of which are specified by the data in the unfolding problem. Let m denote the dimensionality of the Euclidean space into which these points are to be imbedded. If only the linear equations, numbered (18) herein, are to be used, then Schönemann gives the following data requirement for the solution to be uniquely determined:

$$\text{Max } \{P - 1, Q - 1\} \geq m(m + 3)/2.$$

If the full set of linear and nonlinear equations (18–20) are used, then the amount of data required for a solution to be locally unique is relaxed to

$$P + Q - 1 \geq m(m + 3)/2.$$

Both of these results assume that the equations are independent, which has not been proved.

Introduction

Schönemann [1970] has given an algorithm for metric unfolding. The reader is referred to his paper for a discussion of the uses of this type of model and a bibliography.

The object of this paper is to clarify Schönemann's algorithm and, in particular, to make it clear that the equations numbered (3.2) in Schönemann's article, which define Schönemann's solutions, are only part of the set of equations which are obtained by Schönemann's transformation of the original equations which define a metric infolding. Equations (18–20) in this article

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are the complete set of equations. Schönemann uses the equivalent to (18) and omits (19–20) in defining his solutions.

Gordon Bechtel has pointed out to me that this omission has the advantage of omitting the nonlinear equations and retaining a set of linear equations. Schönemann's solutions will include the true unfoldings if any exist. Given infallible data (solution(s) exist), if enough data are available so that there are enough linear equations to uniquely determine a single solution then Schönemann's solution will coincide with the true solution. Let P and Q denote the number of elements in the two sets of points, the interset distances of which are specified by the data in the unfolding problem. Let m denote the dimensionality of the Euclidean space into which these points are to be imbedded. If only the linear equations are to be used, then Schönemann gives the following data requirement for the solution to be uniquely determined:

$$\text{Max } \{P - 1, Q - 1\} \geq m(m + 3)/2. \tag{30}$$

If the full set of linear and nonlinear equations (18–20) are used, then the amount of data required for a solution to be locally unique is relaxed to

$$P + Q - 1 \geq m(m + 3)/2. \tag{29}$$

The data requirement (30) for using only the linear equations (18) assumes that they are linearly independent, and the data requirement (29) of the full set (18–20) assumes that they are locally linearly independent. Neither conjecture has been proved. Using the full set of linear and nonlinear equations, global uniqueness is probably insured by strengthening the requirement of (29) to

$$P + Q - 2 \geq m(m + 3)/2.$$

1. The Metric Unfolding Problem

Consider two finite sets of elements

$$a_i \; ; \quad i = 1, \cdots, P$$

$$b_j \; ; \quad j = 1, \cdots, Q.$$

We are given the distance δ_{ij} from each a_i to each b_j, but we are not given the distances between the a_i's or between the b_j's. The problem is to assign a point in m-dimensional Euclidean space R^m to each a_i and b_j such that the distances between these points in R^m using the usual Euclidean metric will be as specified by the δ_{ij}.

The basic elements of this problem can be written as three matrices, D, A, B: The *square-distance matrix* D is the $P \times Q$ matrix of specified interset distances squared, that is, its ij-th element is

$$D_{ij} = \delta_{ij}^{\,2}. \tag{1}$$

Let any assignment of points in R^m to the a_i's and b_j's be designated by column m-vectors $\mathbf{a}_i$ and $\mathbf{b}_j$. Any such assignment will be called an *imbedding* and will be specified by an $m \times P$ matrix A and an $m \times Q$ matrix B:

$$A = \mathbf{a}_1 \cdots \mathbf{a}_P \qquad B = \mathbf{b}_1 \cdots \mathbf{b}_Q \,. \tag{2}$$

Vectors will always be written as columns.

For any imbedding (A, B) let $D(A, B)$ denote the $P \times Q$ matrix of their interset squared distances:

$$D(A, B)_{ij} = \|\mathbf{a}_i - \mathbf{b}_j\|^2 \,; \tag{3}$$

using this notation, the problem is to find an imbedding (A, B) such that

$$D(A, B) = D. \tag{4}$$

A solution to (4) will be called an *m-dimensional unfolding of D.*

2. *Schönemann's Algorithm*

Let the *row means, column means,* and *total mean* of D be defined by

$$D_{i\cdot} = \mathrm{Ave}_j\, D_{ij} \,; \qquad D_{\cdot j} = \mathrm{Ave}_i\, D_{ij} \,; \qquad D_{\cdot\cdot} = \mathrm{Ave}_{ij}\, D_{ij} \,. \tag{5}$$

Let the doubly centered square-distance matrix C be obtained from D by adding numbers to each row and column of D in such a way that the sum of the elements of any row or column of C is 0. That is,

$$C_{ij} = D_{ij} + D_{\cdot\cdot} - D_{i\cdot} - D_{\cdot j} \,. \tag{6}$$

For any imbedding (A, B) let $D(A, B)_{i\cdot}$, $D(A, B)_{\cdot j}$, $D(A, B)_{\cdot\cdot}$, $C(A, B)$ be derived in the same way from $D(A, B)$:

$$D(A, B)_{i\cdot} = \mathrm{Ave}_j\, D(A, B)_{ij} \qquad D(A, B)_{\cdot j} = \mathrm{Ave}_i\, D(A, B)_{ij} \tag{5'}$$

$$D(A, B)_{\cdot\cdot} = \mathrm{Ave}_{ij}\, D(A, B)_{ij}$$

$$C(A, B)_{ij} = D(A, B)_{ij} + D(A, B)_{\cdot\cdot} - D(A, B)_{i\cdot} - D(A, B)_{\cdot j}. \tag{6'}$$

Then the unfolding problem, expressed by (4), can be rewritten

$$C(A, B) = C \tag{7}$$

$$D(A, B)_{i\cdot} = D_{i\cdot} \,; \qquad i = 1, \cdots, P \tag{8}$$

$$D(A, B)_{\cdot j} = D_{\cdot j} \,; \qquad j = 1, \cdots, Q. \tag{9}$$

Schönemann's algorithm consists of two steps: Step 1) The collection of imbeddings which satisfy (7) is characterized. These imbeddings will be called the *variable mean unfoldings* of D. Step 2) Equations are derived which specify, from among the variable mean unfoldings of D, those which have the correct row and column means, that is, those which satisfy (8, 9).

2.1. Separately mass centered imbeddings.

For any imbedding (A, B), let $\mathbf{a}$ denote the center of mass of the $\mathbf{a}_i$ and $\mathbf{b}$ the center of mass of the $\mathbf{b}_j$:

$$\mathbf{a} = \mathrm{Ave}_i\, \mathbf{a}_i \qquad \mathbf{b} = \mathrm{Ave}_j\, \mathbf{b}_j\,. \tag{10}$$

Let A^* be derived from A by subtracting $\mathbf{a}$ from each of its columns, so that each row of A^* sums to 0, and B^* be similarly derived from B:

$$A^* = A - \overbrace{(\mathbf{a} \cdots \mathbf{a})}^{P \text{ columns}} \qquad B^* = B - \overbrace{(\mathbf{b} \cdots \mathbf{b})}^{Q \text{ columns}}\,. \tag{11}$$

Straightforward algebraic manipulation of the definitions, outlined in Section 5.1, shows that (7) is equivalent to

$$A^{*t}B^* = -C/2. \tag{12}$$

2.2. Rank of imbedding vs rank of C.

An m-dimensional imbedding (A, B) will be called *full rank* if A^* and B^* are each of rank m. Then (12) shows the following:

If $m <$ Rank (C), then there is no m-dimensional unfolding of D.
If $m =$ Rank (C), then any m-dimensional unfolding of D is full rank.
If $m >$ Rank (C), then no m-dimensional unfolding of D is full rank.

If $m =$ Rank (C), then Schönemann gives us the following procedure for finding the unfoldings of D. The case $m >$ Rank (C) is not treated.

2.3. Step 1: The variable mean unfoldings.

Let $m =$ Rank (C). Find any $m \times P$ matrix A_0 and any $m \times Q$ matrix B_0 such that

$$A_0{}^tB_0 = -C/2. \tag{13}$$

It is well known that such a factorization of $-C/2$ must exist, and there are algorithms for obtaining such a factorization. In Section 5.2 it is shown that A_0 and B_0 are necessarily separately mass centered, that is

$$A_0{}^* = A_0 \qquad B_0{}^* = B_0\,. \tag{14}$$

Therefore, (12) is satisfied and $(A_0\,,\, B_0)$ is a prototype variable mean unfolding of D.

In Section 5.3 it is shown that the collection of variable mean unfoldings of D is precisely the collection of imbeddings $(A(T, \mathbf{a}, \mathbf{b}), B(T, \mathbf{a}, \mathbf{b}))$ derived from $(A_0\,,\, B_0)$ as follows, where T ranges over non-singular $m \times m$ matrices and $\mathbf{a}$ and $\mathbf{b}$ range over m-vectors:

$$\begin{aligned} A(T, \mathbf{a}, \mathbf{b}) &= TA_0 + (\mathbf{a} \cdots \mathbf{a}) \\ B(T, \mathbf{a}, \mathbf{b}) &= T^{-1t}B_0 + (\mathbf{b} \cdots \mathbf{b}). \end{aligned} \tag{15}$$

2.4. Rotation-Translation equivalence classes.

For each variable mean unfolding of D characterized by $(T, \mathbf{a}, \mathbf{b})$, define

$$M = T^t T \qquad \mathbf{x} = T^t(\mathbf{a} - \mathbf{b}). \tag{16}$$

It is shown in Section 5.4 that two variable mean unfoldings of D differ by a rotation and translation of R^m if and only if they have the same $(M, \mathbf{x})$. If M is any $m \times m$ symmetric matrix, then it is well known that there exist $m \times m$ non-singular matrices T such that $M = T^t T$ if and only if M is positive definite, and algorithms exist for obtaining such a factorization. Therefore, the rotation-translation equivalence classes of the variable mean unfoldings of D are in one-to-one correspondence with the pairs $(M, \mathbf{x})$, where M is $m \times m$ symmetric positive definite and $\mathbf{x}$ is an m-vector.

The original data, which specifies only interpoint distances, is rotationally and translationally ambiguous concerning the positions of the points. Therefore, it is advantageous to describe a configuration of points by $(M, \mathbf{x})$ rather than by $(T, \mathbf{a}, \mathbf{b})$, since $(M, \mathbf{x})$ describes the configuration up to rotational-translational ambiguity.

2.5. Step 2: The fixed mean unfoldings.

If the formulae (15) for the variable mean unfoldings of D are inserted into the formula (3) for the square distance matrix of an imbedding, the resulting square distance-matrix $D(T, \mathbf{a}, \mathbf{b})$ is found to depend only on $(M, \mathbf{x})$, as is to be expected. This computation is displayed in Section 5.5. Therefore, the left sides of the row and column mean (8, 9) only depend on $(M, \mathbf{x})$. These constraints on $(M, \mathbf{x})$ are written out explicitly in (18–20) below, which are derived in Section 5.6.

Thus the rotation-translation equivalence classes of unfoldings of D are in one-to-one correspondence with the pairs $(M, \mathbf{x})$ such that M is $m \times m$ symmetric positive definite, $\mathbf{x}$ is an m-vector and $(M, \mathbf{x})$ satisfies (18–20). Given any such solution $(M, \mathbf{x})$ to (18–20), a representative $(A(T, \mathbf{a}, \mathbf{b}), B(T, \mathbf{a}, \mathbf{b}))$ of the corresponding rotation-translation equivalence class of unfoldings of D can be found by factoring M into $T^t T$, determining $\mathbf{a} - \mathbf{b}$ by the second of (16), choosing any $\mathbf{a}$ and $\mathbf{b}$ with this difference, and substituting the triple $(T, \mathbf{a}, \mathbf{b})$ thus found into (15).

The $P + Q$ equations (8, 9) have a dependency: The average of (8) is the same as the average of (9), both being

$$D(A, B)_{\cdot\cdot} = D_{\cdot\cdot}\,. \tag{17}$$

Therefore, any one of these equations can be eliminated, leaving $P + Q - 1$ presumably independent equations. These are expressed most simply by subtracting the last of (8) from the previous $P - 1$ equations and the last of (9) from the previous $Q - 1$ equations, and including (17). Substituting

the formulae (15) for the variable mean unfoldings of D into these equations gives

$$(18)\quad \mathbf{a}_{0i}{}^t M\mathbf{a}_{0i} - \mathbf{a}_{0P}{}^t M\mathbf{a}_{0P} - 2(\mathbf{a}_{0i} - \mathbf{a}_{0P})^t\mathbf{x} = D_{i\cdot} - D_{P\cdot}\,;\, i = 1, \cdots, P-1;$$

$$(19)\quad \mathbf{b}_{0j}{}^t M^{-1}\mathbf{b}_{0j} - \mathbf{b}_{0Q}{}^t M^{-1}\mathbf{b}_{0Q} - 2(\mathbf{b}_{0j} - \mathbf{b}_{0Q})^t M^{-1}\mathbf{x} = D_{\cdot j} - D_{\cdot Q}\,;\, j = 1, \cdots, Q-1;$$

$$(20)\quad \mathrm{Ave}_i\, \mathbf{a}_{0i}{}^t M\mathbf{a}_{0i} + \mathrm{Ave}_j \mathbf{b}_{0j}{}^t M^{-1}\mathbf{b}_{0j} + \mathbf{x}^t M^{-1}\mathbf{x} = D_{\cdot\cdot}\,.$$

Thus, the metric unfolding problem is reduced to $P + Q - 1$ equations in $m(m + 3)/2$ unknowns, namely, the m components of $\mathbf{x}$ and the $m(m + 1)/2$ independent components of the symmetric matrix M.

The reason for comparing the number of degrees of freedom in choosing an imbedding to the number of restraints imposed by the data is in order to determine how much data is required to uniquely determine the proper imbedding. The above comparison was based on Schönemann's transformation, expressed by (7–9). Namely, $m(m + 3)/2$ is the number of degrees of freedom in choosing a rotation-translation equivalence class of imbeddings (A, B) with specified $C(A, B)$. $P + Q - 1$ is the number of restraints imposed on these degrees of freedom by (8, 9).

Of course the difference between the number of degrees of freedom in choosing a rotation-translation equivalence class of imbeddings and the number of restraints imposed by the data can be determined from the original definition (4) of an unfolding without using Schönemann's transformation to reduce both numbers by the number of degrees of freedom of $C(A, B)$. This is done as follows, the results of both methods of analysis being summarized in Table 1.

In the original formulation of the problem, imbedding $P + Q$ elements in R^m gave us $(P + Q)m$ degrees of freedom. It would appear that the $P \times Q$ square-distance matrix D imposes PQ constraints on the allowable imbeddings, but the elements of D are not independent if a full rank m-dimensional unfolding is to be possible. Namely, C must be rank m. The total number of independent constraints imposed on the unfoldings by the data equals the number of degrees of freedom in the data D subject to the constraint that C be of rank m. The number of degrees of freedom in the data equals the number

TABLE 1.

The number of constraints imposed on a metric unfolding by the data is compared to the number of degrees of freedom in choosing a rotation-translation equivalence class of imbeddings. Two derivations are shown which give the same result. Namely, in the first column the number of constraints imposed by the entire data set is compared to the total number of degrees of freedom, using the original definition of metric unfolding. The second column is derived via Schönemann's transformation after removing the degrees of freedom in $C(A, B) = C$. That is, the imbedding (A, B) is allowed to vary only in ways which change the row and column means of $D(A, B)$, with C(A, B) fixed.

TABLE 1

	DEGREES OF FREEDOM		
	(A,B) to be determined, C is of rank m	Schönemann's transform: C(A,B)=C fixed, (T,a,b) to be determined	Difference = No. degrees freedom in C(A,B)=C of rank m
TOTAL No. degrees of freedom in choosing an imbedding	m(P+Q)	m^2+2m	m(P+Q-2)
No. rotational-translational degrees of freedom	m(m+1)/2	m(m+1)/2	--------
Difference = No. degrees freedom in rotational-translational equivalence class	m(2P+2Q-m-1)/2	m(m+3)/2	m(P+Q-2)
No. independent constraints by eqs(7-9) in column 1, by eqs(18-20) in column 2	P+Q-1 + m(P+Q-m-2)	P+Q-1	--------
Difference = excess No. of constraints (over No. of degrees of freedom)	m(m+3)/2 - (P+Q-1)	m(m+3)/2 - (P+Q-1)	--------

of degrees of freedom in the row and column means of D plus the number of degrees of freedom in a $P \times Q$ matrix C of rank m. There are $P + Q - 1$ independent row and column means of D. Since the rows and columns of C sum to 0, the last row and column are determined by the remaining $(P - 1) \times (Q - 1)$ matrix which must be of rank m. Its first m rows are independent, giving $m(Q - 1)$ independent components, but its remaining $P - 1 - m$ rows must be linear combinations of the first m rows, and so contain only m independent components each. Thus, expressing D by means of its row and column means and C, we find that the number of independent constraints imposed on its unfoldings is

$$(P + Q - 1) + m(Q - 1) + (P - 1 - m)m. \tag{21}$$

Within each rotation-translation equivalence class of imbeddings there are $m(m - 1)/2$ degrees of rotational freedom and m degrees of translational freedom. Therefore, the number of degrees of freedom in choosing a rotation-translation equivalence class of imbeddings is

$$(P + Q)m - m(m + 1)/2. \tag{22}$$

Subtracting (22) from (21) gives the excess number of constraints imposed by the data, over the number of degrees of freedom in choosing a rotation-translation equivalence class of imbeddings, to be

$$(P + Q - 1) - m(m + 3)/2.$$

Suppose that D consists of fallible data, that is, it doesn't necessarily possess an unfolding and we are trying to find the imbedding which best fits it. In solving for $(M, \mathbf{x})$ one might obtain an M which is not positive definite. Indeed, choosing M at random this is the more likely situation. In this case M cannot be factored into T^tT. Therefore, it might be necessary to replace M by T^tT in (18–20) and solve for the best $(T, \mathbf{x})$. This expedient has the undesirable effect of unnecessarily increasing the number of variables by the $m(m - 1)/2$ degrees of rotational freedom. These superfluous variables can be eliminated by requiring T to be of some canonical form which can always be obtained by multiplying T by a rotation (orthonormal) matrix. For instance, it is well known that T can be required to be 0 above the principal diagonal. Then it is easy to see that T has the same number of degrees of freedom as the symmetric M. Furthermore, it is easy to determine if T is nonsingular when it has this form, namely, its diagonal elements must differ from 0. However, even if T is singular (which is unlikely), we still obtain an imbedding (unlike the case in which M is not positive semidefinite), it just is not full rank.

3. *Variable Column Means: Linearization*

3.1. *The variable column mean unfolding problem.*

Suppose that the data D_{ij} differs from the square distance $||a_i - b_j||^2$ by an unknown amount e_j which is constant over each column of D. Our problem is to find an imbedding (A, B) such that

$$D(A, B)_{ij} = D_{ij} + e_j \tag{24}$$

for some vector $(e_1, \cdots, e_Q)$ of real numbers. That is, the data specifies $D(A, B)$ up to the column means $D(A, B)_{\cdot j}$ and, therefore, the average $D(A, B)_{\cdot\cdot}$ of the row means $D(A, B)_{i\cdot}$.

In this problem $C(A, B)$ and the differences $D(A, B)_{i\cdot} - D(A, B)_{k\cdot}$ between the row means are specified. Therefore, the previous algorithm can be used, except that the $m(m + 3)/2$ variables $(M, \mathbf{x})$ need only satisfy the $P - 1$ equations (18), which are linear!

In the variable row mean unfolding problem we obtain the $Q - 1$ equations (19). These can be linearized by replacing the variables $(M, \mathbf{x})$ by $(N, \mathbf{y})$:

$$N = M^{-1} \qquad \mathbf{y} = M^{-1}\mathbf{x}. \tag{25}$$

3.2. *The linearized unfolding algorithm.*

In describing his algorithm for unfolding D, Schönemann specifies the $P - 1$ constraints (18) on $(M, \mathbf{x})$, and omits the Q constraints (19, 20). Namely, his (3.2) give the constraints determined by $f_{ij} - f_{pj}$ in his notation, which equals the difference between the ith and the last row mean, and he omits the constraints determined by his f_{pj} for each of the Q values of j.

This omission has the benefit of only leaving linear equations to be solved for $(M, \mathbf{x})$. Gordon Bechtel has pointed out to me that the remaining equations will specify the correct solution if we are given enough data D so that $(M, \mathbf{x})$ is still uniquely determined. Since linearization is possible by omitting the constraints determined by either the column or row means of D, the data requirement for $(M, \mathbf{x})$ to be determined by linear equations is

$$\max \{P - 1, Q - 1\} \geq m(m + 3)/2. \tag{26}$$

4. *Uniqueness of Solution*

Suppose we have a mess of equations $f_r(x_s) = 0$ in variables x_s. Let $\mathbf{x}_0$ be a solution. $\mathbf{x}_0$ will be called *globally unique* if there is no other solution, and *locally unique* if, for some $\epsilon > 0$, there is no other solution in the ϵ-neighborhood of $\mathbf{x}_0$.

$\mathbf{x}_0$ is locally unique if the rank of the Jacobian matrix $(\partial f_r/\partial x_s)$ equals the number of variables, when evaluated at $\mathbf{x}_0$. If the f_r are linear then this condition insures global uniqueness, and if, furthermore, the number of

equations equals the number of variables, then this condition insures that a solution will exist.

In the use of the non-linearized unfolding (18–20) it should be noted that even if the equations are independent and the number of equations equals the numbers of variables in $(M, \mathbf{x})$, *i.e.*,

$$P + Q - 1 = m(m + 3)/2, \tag{27}$$

we are insured neither that a solution exists nor, if it does, that it is globally unique. We are only insured that every solution is locally unique.

In the case $P = Q = 3$, $m = 2$, which satisfies (27), Bill Chaplin has provided me with the example in Figure 1 of two imbeddings with the same interset distances, so that if D consists of these distances squared, then there are at least two solutions. Each of these solutions is locally unique, which is shown by computing the Jacobian of the constraints imposed on the imbedding by the data. For instance, consider the imbedding on the left side of Fig. 1. The translational freedom can be removed by requiring $\mathbf{a}_3$ to be at the origin and the rotational freedom can be removed by requiring $\mathbf{b}_1$ to lie on the horizontal axis passing through $\mathbf{a}_3$. This leaves one horizontal degree of freedom in choosing $\mathbf{b}_1$, and two degrees of freedom each in choosing $\mathbf{a}_1$, $\mathbf{a}_2$, $\mathbf{b}_2$, $\mathbf{b}_3$, a total of 9 degrees of freedom in choosing the imbedding. The data imposes 9 constraints on this imbedding, namely, the square-distance $||\mathbf{a}_i - \mathbf{b}_j||^2$ is specified for $i, j = 1, 2, 3$. Table 2 shows the 9×9 Jacobian matrix of the constraints, obtained by taking the derivative of each constraint with respect to each degree of freedom in choosing the imbedding, these

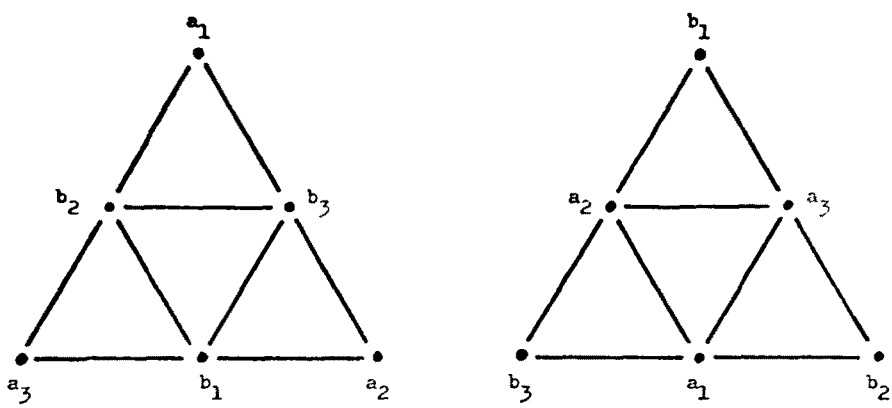

FIGURE 1.

Two imbeddings with the same interset distances for the case $P = Q = 3$, $m = 2$, which satisfies (27).

TABLE 2.

Partial derivatives of the interpoint distances in Fig. 1 with respect to displacements of the points. This matrix is nonsingular, showing that the configuration of Fig. 1 is a locally unique solution for its interpoint distances.

TABLE 2

	b_1 →	b_2 →	b_2 ↑	b_3 →	b_3 ↑	a_1 →	a_1 ↑	a_2 →	a_2 ↑
$\|a_1-b_1\|^2$	0	0	0	0	0	0	$2\sqrt{3}$	0	0
$\|a_1-b_2\|^2$	0	-1	$-\sqrt{3}$	0	0	1	$\sqrt{3}$	0	0
$\|a_1-b_3\|^2$	0	0	0	1	$-\sqrt{3}$	-1	$\sqrt{3}$	0	0
$\|a_2-b_1\|^2$	-2	0	0	0	0	0	0	2	0
$\|a_2-b_2\|^2$	0	-3	$\sqrt{3}$	0	0	0	0	3	$-\sqrt{3}$
$\|a_2-b_3\|^2$	0	0	0	-1	$\sqrt{3}$	0	0	1	$-\sqrt{3}$
$\|a_3-b_1\|^2$	2	0	0	0	0	0	0	0	0
$\|a_3-b_2\|^2$	0	1	$\sqrt{3}$	0	0	0	0	0	0
$\|a_3-b_3\|^2$	0	0	0	3	$\sqrt{3}$	0	0	0	0

derivatives being evaluated at the particular imbedding shown on the left side of Fig. 1. Each row is labelled by a constraint and each colunn by a degree of freedom in the choice of the imbedding. This matrix is easily shown to be nonsingular by successively adding a scalar times one column to another column in such a way as to change nonzero elements to zero, until each column has exactly one nonzero element and these are on different rows. Since the imbedding on the left side of Fig. 1 is assumed to satisfy the constraints, it follows that any small perturbation of the imbedding will fail to satisfy the constraints.

Thus, local uniqueness does not necessarily imply global uniqueness in the case of nonlinear equations. There are two reasons for discussing local uniqueness here: Firstly, if the data is insufficient to provide a locally unique solution, then it is certainly insufficient to provide a globally unique solution. That is, local uniqueness is a necessary condition for global uniqueness. Secondly, if the data is sufficient to guarantee local uniqueness, then the solutions must be discrete, *i.e.* separated by finite distances. It seems likely that increasing the data to provide one more constraint on the solutions will determine the solution uniquely.

In summary, assuming Rank $(C) = m$, and assuming that (18–20) are independent, we obtain the following data requirements for unique unfolding:

1) In the variable mean unfolding problem, $D(A, B)$ is specified up to constants to be added to its rows and columns, *i.e.*, $C(A, B)$ is specified but not the row or column means of $D(A, B)$. In this problem, Rank $(C) = m$ insures that a solution exists, and the solution is neither locally nor globally unique. The solutions are specified by (15).

2) In the variable column mean unfolding problem, in which the column means of $D(A, B)$ are not specified but the differences between the row means are specified, a solution is globally unique if

$$P - 1 \geq m(m + 3)/2. \tag{28}$$

Furthermore, a solution to (18) is guaranteed if equality holds, but M may not be positive definite, so that there may be no solution to the unfolding problem.

3) In the fixed mean unfolding problem, in which D is completely specified, we are again not guaranteed a solution. But, if a solution exists, then

$$P + Q - 1 \geq m(m + 3)/2 \tag{29}$$

guarantees local uniqueness. Presumably, the strict inequality will insure global uniqueness.

4) In the fixed mean unfolding problem, if we only wish to use a subset of (18–20) which are linear, then

$$\max \{P - 1, Q - 1\} \geq m(m + 3)/2 \tag{30}$$

insures a globally unique solution, if it exists.

5. *Proofs*

The following proofs are primarily due to Schönemann. They are presented here in my notation for clarification.

5.1. Proof that the variable mean unfoldings are defined by $A^{*t}B^* = -C/2$ (*equation* (12)).

The variable mean unfoldings were defined by (7) to be those imbeddings (A, B) such that $C(A, B) = C$. By definitions (6′) and (3),

$$\begin{aligned} C(A, B)_{ij} &= D(A, B)_{ij} \\ &\quad + \mathrm{Ave}_{ij}\, D(A, B)_{ij} - \mathrm{Ave}_i\, D(A, B)_{ij} - \mathrm{Ave}_j\, D(A, B)_{ij} \end{aligned} \tag{31}$$

$$D(A, B)_{ij} = ||\mathbf{a}_i - \mathbf{b}_j||^2 = ||\mathbf{a}_i||^2 - 2\mathbf{a}_i{}^t\mathbf{b}_j + ||\mathbf{b}_j||^2. \tag{32}$$

Substituting (32) into (31) gives

$$C(A, B)_{ij} = -2(\mathbf{a}_i - \mathbf{a})^t(\mathbf{b}_j - \mathbf{b}) = -2\mathbf{a}_i{}^{*t}\mathbf{b}_j{}^*, \tag{33}$$

where

$$\mathbf{a} = \mathrm{Ave}_i\, \mathbf{a}_i \qquad \mathbf{b} = \mathrm{Ave}_j\, \mathbf{b}_j\,. \tag{34}$$

5.2. Proof that A_0 and B_0 are separately mass centered.

By definition (13),

$$A_0{}^t B_0 = -C/2.$$

C was constructed so that the sum of the elements of each of its rows is 0, that is, $C\mathbf{1} = \mathbf{0}$, where $\mathbf{1}$ is a column of 1's. Therefore, $A_0{}^t$ annihilates the m-vector $B_0\mathbf{1}$. Since A_0 is of rank m, this implies $B_0\mathbf{1} = \mathbf{0}$ that is, the sum of the columns of B_0 is $\mathbf{0}$. Similarly, the fact that the sum of the elements of each column of C is 0 implies that the sum of the columns of A_0 is $\mathbf{0}$.

5.3. Proof that $\{(A(T, \mathbf{a}, \mathbf{b}), B(T, \mathbf{a}, \mathbf{b}))\}$ is precisely the set of variable mean unfoldings.

In order to see that every $(A(T, \mathbf{a}, \mathbf{b}), B(T, \mathbf{a}, \mathbf{b}))$ is a variable mean unfolding of D, note that, since the sum of the columns of A_0 and the sum of the columns of B_0 are $\mathbf{0}$, *i.e.*, each annihilates the column vector of 1's of appropriate length, the same is true of TA_0 and $T^{-1t}B_0$. Therefore,

$$A(T, \mathbf{a}, \mathbf{b})^* = (TA_0 + \mathbf{a} \cdots \mathbf{a})^* = TA_0$$

$$B(T, \mathbf{a}, \mathbf{b})^* = (T^{-1t}B_0 + \mathbf{b} \cdots \mathbf{b})^* = T^{-1t}B_0\,, \tag{35}$$

so that

$$A(T, \mathbf{a}, \mathbf{b})^{*t} B(T, \mathbf{a}, \mathbf{b})^* = A_0{}^t B_0 = -C/2$$

which is condition (12), which was proven in the Section 5.1 to be equivalent to the definition of a variable mean unfolding.

To see that any variable mean unfolding (A, B) of D is of this form, note first that if we take $\mathbf{a}$ to be the average of the columns of A and $\mathbf{b}$ to be the average of the columns of B, then

$$A = A^* + \mathbf{a} \cdots \mathbf{a} \qquad B = B^* + \mathbf{b} \cdots \mathbf{b} \tag{36}$$

and second that, from condition (12) and definition (13),

$$A^{*t} B^* = -C/2 = A_0{}^t B_0\,. \tag{37}$$

That is, (A^{*t}, B^*) and $(A_0{}^t, B_0)$ are full rank factorizations of the matrix $-C/2$. It is well-known that this implies that, for some nonsingular T,

$$A^* = TA_0 \qquad B^* = T^{-1t}B_0\,. \tag{38}$$

Equations (36, 38) show that (A, B) is of the form (15).

5.4. Proof that two variable mean unfoldings of D are rotation-translation equivalent iff they have the same $(M, \mathbf{x})$.

To say that two imbeddings (A, B) and (A', B') are *rotation-translation equivalent* means that there is a rotation R of R^m and an m-vector $\mathbf{t}$ such that

$$A' = RA + \mathbf{t} \cdots \mathbf{t} \qquad B' = RB + \mathbf{t} \cdots \mathbf{t}. \tag{39}$$

Therefore, if two variable mean unfoldings $(T, \mathbf{a}, \mathbf{b})$ and $(T', \mathbf{a}', \mathbf{b}')$ of D are rotation-translation equivalent, then definition (15) implies

$$\begin{aligned} T'A_0 + \mathbf{a}' \cdots \mathbf{a}' &= RTA_0 + (R\mathbf{a} + \mathbf{t}) \cdots (R\mathbf{a} + \mathbf{t}) \\ T'^{-1t}B_0 + \mathbf{b}' \cdots \mathbf{b}' &= RT^{-1t}B_0 + (R\mathbf{b} + \mathbf{t}) \cdots (R\mathbf{b} + \mathbf{t}) \end{aligned} \tag{40}$$

for some $(R, \mathbf{t})$. Note that, since the sum of the columns of A_0 is $\mathbf{0}$, the same holds for $T'A_0$ and RTA_0 . Therefore, averaging the columns of each side of (40) gives

$$\mathbf{a}' = R\mathbf{a} + \mathbf{t} \qquad \mathbf{b}' = R\mathbf{b} + \mathbf{t}. \tag{41}$$

Subtracting (41) from (40) gives

$$(T' - RT)A_0 = 0. \tag{42}$$

Since the rank of A_0 is m, (42) shows that $(T' - RT)$ annihilates all of R^m, that is,

$$T' = RT. \tag{43}$$

Equations (43) and (41) now give the desired result:

$$\begin{aligned} M' &= T'^tT' = (RT)^t(RT) = T^tT = M \\ \mathbf{x}' &= T'^t(\mathbf{a}' - \mathbf{b}') = (RT)^t(R\mathbf{a} - R\mathbf{b}) = T^t(\mathbf{a} - \mathbf{b}) = \mathbf{x}. \end{aligned} \tag{44}$$

Conversely, suppose $(M, \mathbf{x}) = (M', \mathbf{x}')$, that is

$$T'^tT' = T^tT \tag{45}$$

$$T'^t(\mathbf{a}' - \mathbf{b}') = T^t(\mathbf{a} - \mathbf{b}). \tag{46}$$

It is a well-known result that (45) implies

$$T' = RT \tag{47}$$

for some rotation R of R^m. Define

$$\mathbf{t} = \mathbf{a}' - R\mathbf{a}. \tag{48}$$

Substituting (47) into (46) yields

$$\mathbf{b}' - R\mathbf{b} = \mathbf{a}' - R\mathbf{a} = \mathbf{t}. \tag{49}$$

Equations (47, 49) give (40), *i.e.*, the two variable mean unfoldings are rotation-translation equivalent.

5.5. Derivation of $D(M, \mathbf{x})$.

Substitution of the definition (15) of $(A(T, \mathbf{a}, \mathbf{b}), B(T, \mathbf{a}, \mathbf{b}))$ into the definition (3) of $D(A, B)$ yields the fundamental equation

$$
\begin{aligned}
D(M, \mathbf{x})_{ij} &= D(T, \mathbf{a}, \mathbf{b})_{ij} && \text{(Def. of } D(M, \mathbf{x})) \\
&= ||\mathbf{a}_i(T, \mathbf{a}, \mathbf{b}) - \mathbf{b}_j(T, \mathbf{a}, \mathbf{b})||^2 && \text{(from (3))} \\
&= ||(T\mathbf{a}_{0i} + \mathbf{a}) - (T^{-1t}\mathbf{b}_{0j} + \mathbf{b})||^2 && \text{(from (15))} \\
&= ||T\mathbf{a}_{0i} - T^{-1t}\mathbf{b}_{0j} + T^{-1t}\mathbf{x}||^2 && \text{(from the second of (16))} \\
&= \mathbf{a}_{0i}{}^tM\mathbf{a}_{0i} + \mathbf{b}_{0j}{}^tM^{-1}\mathbf{b}_{0j} + \mathbf{x}M^{-1}\mathbf{x} \\
&\quad - 2\mathbf{a}_{0i}{}^t\mathbf{b}_{0j} + 2\mathbf{a}_{0i}{}^t\mathbf{x} - 2\mathbf{b}_{0j}{}^tM^{-1}\mathbf{x} && \text{(from the first of (16)).}
\end{aligned} \tag{50}
$$

5.6. Derivation of the row and column means of $D(M, \mathbf{x})$.

Equation (50) can be substituted into the definitions (5′) of the means of a square-distance matrix, which can be substituted into (8, 9) to yield (18–20) after making use of the fact that A_0 and B_0 are separately mass centered, that is

$$\sum_i \mathbf{a}_{0i} = \mathbf{0} \qquad \sum_j \mathbf{b}_{0j} = \mathbf{0}. \tag{51}$$

Substitution of (50) in (5′) gives

$$
\begin{aligned}
D(M, \mathbf{x})_{i.} &= \mathrm{Ave}_j\, D(M, \mathbf{x})_{ij} && \text{(from (5′))} \\
&= \mathbf{a}_{0i}{}^tM\mathbf{a}_{0i} + \mathrm{Ave}_j\, \mathbf{b}_{0j}{}^tM^{-1}\mathbf{b}_{0j} + \mathbf{x}^tM^{-1}\mathbf{x} + 2\mathbf{a}_{0i}{}^t\mathbf{x} \\
& && \text{(from (50, 51))}
\end{aligned} \tag{52}
$$

$$
\begin{aligned}
D(M, \mathbf{x})_{.j} &= \mathrm{Ave}_i\, D(M, \mathbf{x})_{ij} \\
&= \mathrm{Ave}_i\, \mathbf{a}_{0i}{}^tM\mathbf{a}_{0i} + \mathbf{b}_{0j}{}^tM^{-1}\mathbf{b}_{0j} + \mathbf{x}^tM^{-1}\mathbf{x} - 2\mathbf{b}_{0j}{}^tM^{-1}\mathbf{x}
\end{aligned} \tag{53}
$$

$$
\begin{aligned}
D(M, \mathbf{x})_{..} &= \mathrm{Ave}_i\, D(M, \mathbf{x})_{i.} && \text{(from (5′))} \\
&= \mathrm{Ave}_i\, \mathbf{a}_{0i}{}^tM\mathbf{a}_{0i} + \mathrm{Ave}_j\, \mathbf{b}_{0j}{}^tM^{-1}\mathbf{b}_{0j} + \mathbf{x}^tM^{-1}\mathbf{x} \\
& && \text{(from (52, 51)).}
\end{aligned} \tag{54}
$$

Substitution of (52–54) in (8, 9) yields (18–20). For instance, in deriving (18) from (8, 52) note that the second and third terms of (52) do not depend on the index of $D(M, \mathbf{x})_{i.}$ and so drop out of the difference $D(M, \mathbf{x})_{i.} - D(M, \mathbf{x})_{P.}$.

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